

PROFESSIONAL EXPERIENCE

Data Engineer, [Siam Services Pvt Ltd](#) (SSIPL), Remote

Jan 24 – Dec 24

- **Reduced data processing time by 35% by architecting end-to-end ETL pipelines in Python and SQL** to consolidate hydrocarbon well performance and reservoir measurement datasets across SSIPL's Oil & Gas client base, enabling automated scheduling that eliminated **12 hours** of manual data preparation weekly.
- **Engineered Snowflake and dbt-based warehouse models with schema design and incremental transformations** processing liquid level, dynamometer, and pressure transient datasets from SSIPL's well monitoring systems, cutting dashboard load time from **8 to 3 seconds** and enabling near-real-time operational visibility for field engineers.
- **Improved reporting accuracy from 84% to 97% by building automated Python and SQL data quality validation frameworks** aligned with SSIPL's ISO 9001:2015 compliance standards, directly reducing erroneous inputs into production optimization models and saving 6 hours of weekly reconciliation effort.
- **Achieved 99.8% migration accuracy across 3 million historical well test and pressure buildup records** by implementing structured **AWS S3** ingestion pipelines with schema controls and validation checks, reducing on-premises storage costs by **22%** and delivering clean datasets to **Power BI** dashboards used by SSIPL's engineering teams.

Cloud & Data Analytics Intern, [Codelattice Digital Solutions](#), Dubai, UAE

Jun 23 - Aug 23

- **Accelerated client reporting turnaround by 38% by building automated Python and SQL data pipelines with dbt transformation logic** consolidating multi-source campaign and engagement data across Codelattice's AWS hosted client platforms, enabling reliable weekly analytics delivery to over **15 client accounts** in the Dubai market.
- **Designed Google BigQuery and Dataflow analytical models with Power BI dashboards** tracking customer sentiment and engagement KPIs for Codelattice's Reviewlattice product, improving data refresh frequency from daily to **under 90 seconds** and reducing manual reporting effort by **5 hours weekly** for the analytics team.

Data Analytics Intern (using Deep Learning), [Hewlett-Packard Enterprise](#), Singapore (@ NUS)

Jun 22 – Jul 22

- **Engineered a CNN-based image classification model using TensorFlow, Keras, and Scikit-learn** as the capstone project for HPE's certified Deep Learning program at NUS Singapore, achieving **91% validation accuracy** through ResNet transfer learning, hyperparameter tuning, and dropout regularization across structured training and test datasets.
- **Reduced EDA and data preparation turnaround by 40% by building reusable Python, Pandas, and NumPy preprocessing pipelines** for multi-format datasets, enabling faster model ready feature engineering and cutting manual cleaning effort by **8 hours** across the full project lifecycle, evaluated and certified by HPE Education Asia Pacific.

PROJECT EXPERIENCE

HealthPulse: Real-Time Clinical Trial Analytics Lakehouse

- **Engineered a real-time ingestion pipeline using Apache Kafka and PySpark on Databricks** to process clinical trial event streams into a **Delta Lake** Lakehouse architecture, reducing data latency from batch level hours to **under 4 minutes** and enabling continuous trial monitoring across **3 research grade clinical datasets**.
- **Accelerated trial performance reporting by 38% by designing Azure Data Factory** orchestration workflows and **dbt**-modeled transformation layers feeding **Tableau** dashboards, delivering patient enrollment trends, adverse event tracking, and site-level KPIs across structured and semi-structured clinical data sources.

MarketMind: AI Powered Supply Chain Intelligence Engine

- **Built a RAG pipeline using LangChain and MLflow containerized via Docker**, enabling natural language querying over supply chain disruption datasets and cutting manual analyst research time by **40%** by surfacing contextually accurate procurement insights from **~350,000** indexed vendor and inventory records.
- **Improved pipeline data reliability by 34% by implementing Soda Core quality checks** embedded across **Snowflake** ingestion layers monitoring freshness, schema drift, and anomaly detection across vendor feeds, with findings visualized through an interactive **Streamlit** dashboard for operations teams.

EDUCATION

Northeastern University, Boston, MA

Jan 25 - May 27

Masters in Data Analytics Engineering

3.8/4 GPA

Relevant Coursework: Data Mining, Computation & Visualization for Analytics, Database Management Systems, Neural Networks & Deep Learning

BITS Pilani Dubai Campus, Dubai, UAE

Bachelor of Engineering in Computer Science (Graduated with Distinction - top 1%)

Sep 20 – Jul 24

Relevant Coursework: Data Structures & Algorithms, OOP, Database Management Systems, Operating Systems, Computer Networks, Machine Learning & AI, Deep Learning

SKILLS, TOOLS & CERTIFICATIONS

Skills: Data Engineering, ETL/ELT Pipeline Development, Data Modeling, Hypothesis Testing, A/B Testing, Regression Analysis, EDA, Data Warehousing, Lakehouse Architecture, Real-Time Streaming, Data Quality & Observability, Data Lineage, Machine Learning, Deep Learning, NLP, RAG Pipelines, Data Storytelling, Dashboarding & Reporting, Agile/Scrum

Tools: Python, SQL, R, PySpark, Pandas, NumPy, Scikit-learn, Apache Kafka, Apache Spark, Databricks, Delta Lake, dbt, Snowflake, BigQuery, AWS S3, AWS Glue, Azure Data Factory, Google Dataflow, Airflow, LangChain, MLflow, Docker, Soda Core, Tableau, Power BI, Looker, Streamlit, TensorFlow, Keras, PostgreSQL, MySQL, Excel (Power Query, PivotTables), Git, GitHub, CI/CD, JIRA

Certifications (In Progress): AWS Certified Data Engineer Associate, Microsoft Power BI Data Analyst Associate (PL-300), Tableau Certified Data Analyst, Databricks Certified Generative AI Engineer Associate, Google Professional Data Engineer